CLOUD & DEVOPS

Multi-Region Architecture Visual Guide

Active-active vs active-passive — design for distance and disaster.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Why Multiple Regions

Three reasons, three different designs.

Latency

serve users from nearby — 200ms → 30ms

Resilience

survive a regional outage — no single point of geography

Compliance

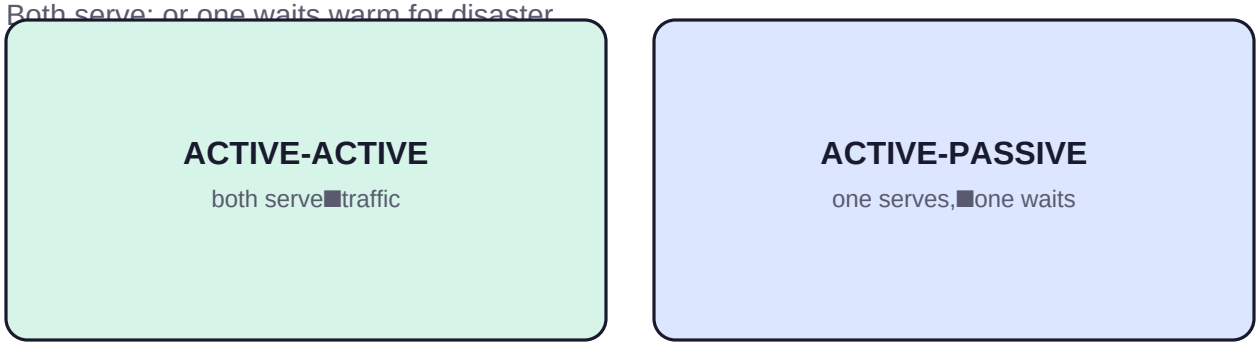
data residency: EU data stays in the EU

Multi-region is the most expensive reliability you can buy — earn it with real requirements, not vibes.

STEP 2/6

Active-Active vs Active-Passive

Both serve; or one waits warm for disaster.



- 2x cost, always
- Needs conflict resolution
- Zero failover time

- ~1.2x cost (warm standby)
- Simpler data story
- Minutes to fail over

Active-active demands multi-writer data. If your database can't do it, you're active-passive with extra steps.

STEP 3/6

Data Replication Patterns

Reads scale easily; writes are the hard part.

Read replicas

one writer, many regional readers — async lag in seconds

Multi-writer

writes anywhere — needs conflict resolution (CRDTs / last-write-wins)

Partitioned

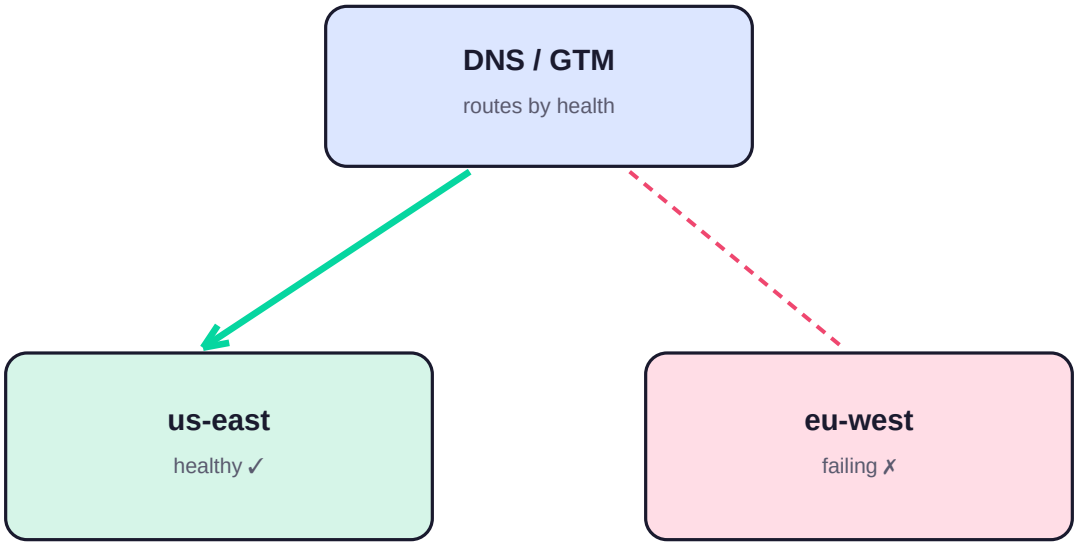
EU users → EU writes; US → US — no conflicts by design

Async replication lag means a reader can see stale data — design the UX for it (or pay for sync).

STEP 4/6

DNS Failover & Health Checks

Traffic steering is only as good as the health signal.



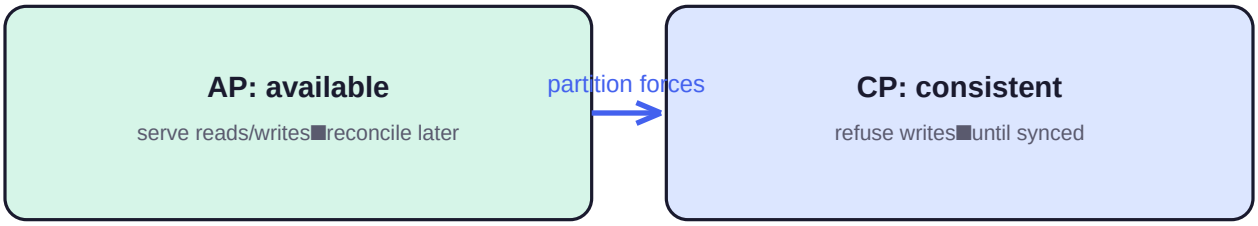
Health checks must test the FULL path: LB → app → database. A 200 from the LB means nothing.

DNS TTL is your failover floor: 60s TTL ≈ 60s+ for clients to move. Anycast + LB failover is faster.

STEP 5/6

Consistency Trade-offs (CAP)

In a partition, pick: availability or consistency.



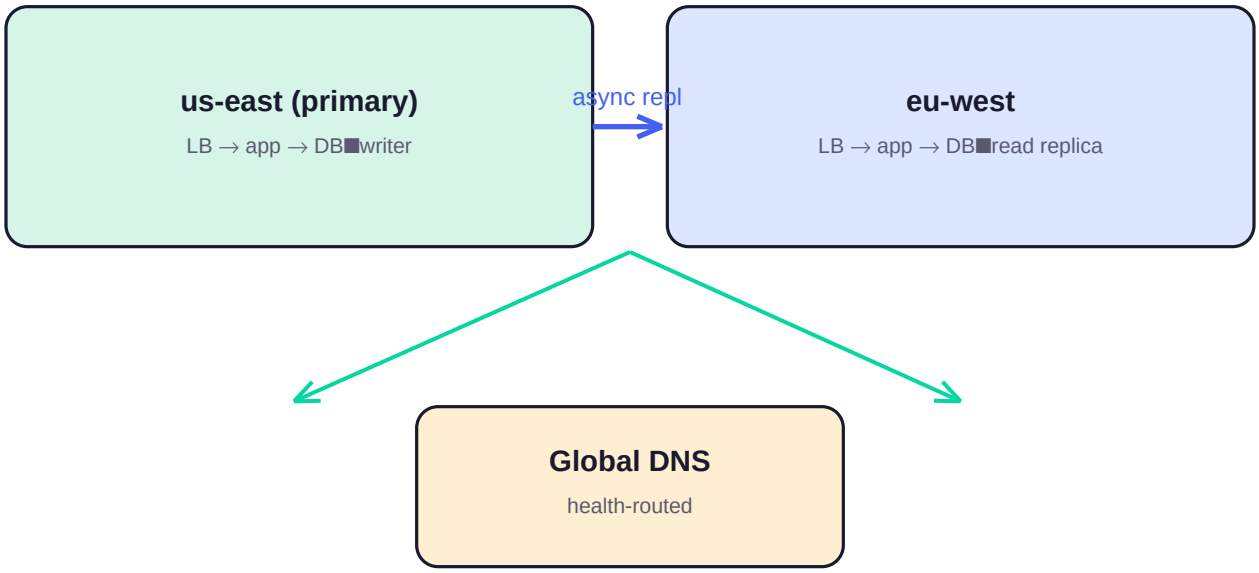
Most apps are mixed: AP for the catalog (stale OK), CP for payments (never wrong).

- Eventual consistency needs a UX: 'order received' now, 'confirmed' after sync.
- Idempotency keys make retried writes safe across regions.
- Test partitions with chaos experiments — don't discover CAP during a real outage.

STEP 6/6

Reference: Regional 3-Tier

Each region is a full stack; DNS + replication tie them together.



Start here: active-passive with read replicas. Go active-active only when writes demand it.

STEP ✓/✓

Key Takeaways

Multi-Region Architecture — Remember

- Go multi-region for latency, resilience, or compliance — never by default.
- Active-active needs multi-writer data; active-passive is cheaper and simpler.
- Replicate reads easily; writes need conflict strategy or partitioning.
- Failover rides on honest health checks and low DNS TTLs.
- CAP forces a choice per workload: AP for catalogs, CP for money.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev